Sensor applications

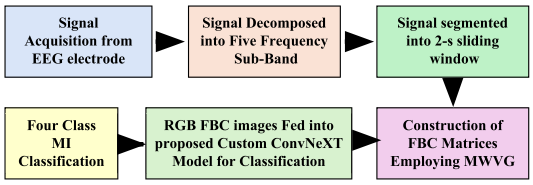
A Novel Multilayer Functional Brain Connectivity-Based Motor Imagery Classification Model Using EEG Sensor Data

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Abstract—This letter presents a novel electroencephalography (EEG) rhythm-based motor imagery (MI) classification framework employing multilayer weighted visibility graph (WVG) and deep learning. In this study, multichannel EEG signals corresponding to different MI classes were acquired from various subjects by placing sensors on the scalp. The acquired EEG signals were initially decomposed into five frequency subbands (EEG rhythms), by segmenting into 2-s overlapping windows. For each rhythm, a multilayer functional brain network has been constructed utilizing WVG, and the resulting functional brain connectivity matrices were mapped into RGB images, which were further classified through a lightweight custom ConvNeXt model. In this work, subject independent evaluation was performed using a leave-one-subject-out protocol with stratified tenfold cross-validation. Experimental validation on two datasets, BCI Competition IV-2a and High Gamma Dataset (HGD), yielded accuracies of 90.20% and 96.1%, respectively. The analysis reveals physiological meaningful connectivity patterns, such as contralateral β-band connectivity in BCI IV-2a and enhanced interhemispheric γ-band integration in HGD. Ablation studies and benchmark comparison confirmed that the proposed framework achieved high classification accuracy, demonstrating the efficiency and potential for robust, real-time subject-independent MI–BCI applications.



Index Terms—Sensor applications, brain–computer interface (BCI), brain connectivity, brain rhythms, classification, deep learning, electroencephalography (EEG) and visibility graph (VG).

I. INTRODUCTION

Brain–computer interface (BCI) systems convert neural activity into executable commands for applications in neurorehabilitation, assistive technologies, and advanced human–machine interaction (HMI), including robotics and prosthetic control [1]. Among various paradigms, motor imagery (MI) enables users to generate control signals by imagining specific limb movements without execution. Electroencephalography (EEG) is widely used in MI–BCI due to its noninvasiveness and high temporal resolution. However, decoding of MI from EEG remains challenging due to low signal-to-noise ratio, nonstationarity, and high intersubject variability.

Recent studies have explored several EEG-based techniques coupled with deep learning models for MI decoding [1], [2], [3], [4], [5], [6], [7], [8], [9]. A physics-informed temporal convolutional network was reported in [2], while Hou et al. [3] introduced a graph convolutional neural network (CNN) for EEG-based MI classification. A channel-projection mixed-scale CNN-based spatial–spectral representation framework was reported in [4] for MI classification. In [5], a combined attention–inception with long short-term memory (LSTM) was reported for rehabilitation-oriented MI decoding. In [6], a CNN model with sliding windows, and in [7], a multiscale spatiotemporal feature fusion neural network for classification of MI-related BCI have been successfully employed. Applications of multiscale CNN [8] and adaptive transfer learning-based multiscale feature fused deep CNN [9] for MI classification has been reported.

Despite substantial progress in deep learning-based MI–EEG decoding, robust subject-independent performance remains challenging due to the intersubject variability and inadequate exploitation of rhythm-wise functional interactions across brain regions. Further, mapping of EEG-rhythms across different channels to construct functional-brain connectivity (FBC) remains unexplored. These limitations motivate us to develop a lightweight EEG rhythm-based brain mapping framework using multilayer weighted visibility graph (MWVG) for subject-independent MI classification. The main contributions of this work are as follows.

1) A WVG-aided multilayer FBC representation has been proposed to analyze EEG rhythm-specific connectivity for MI tasks.
2) A next-generation lightweight custom ConvNeXt model with reduced parameters was implemented for subject-independent MI classification.
3) Extensive ablation studies and benchmark comparison with other deep learning models were conducted on two publicly available datasets.

II. EEG DATA ACQUISITION AND PREPROCESSING

A. EEG Data Acquisition

1) Dataset I: It contains EEG recordings FR 9 participants obtained from BCI competition IV dataset 2a [10]. Participants performed

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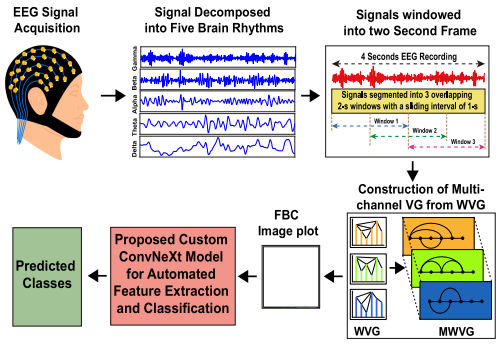


Fig. 1. Flowchart of the proposed MI–BCI classification framework.

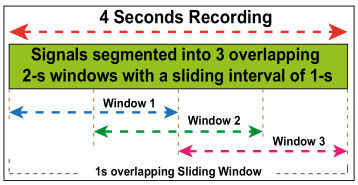


Fig. 2. Graphical illustration of sliding window segmentation.

four-class cue-based MI tasks (left hand, right hand, feet, and tongue). Each subject completed two sessions consisting of 288 trials. EEG signals acquired from 22 scalp electrodes with sampling frequency of 250 Hz. In addition to EEG, EOG channel was provided but were not used in our analysis. Signals were sampled at 250 Hz, with each trail lasting 4 s.

2) Dataset II: The dataset II (HGD) includes EEG data recorded from 14 subjects performing a four class arrow cues MI task (left hand, right hand, both feet, and rest [11]. Each subject completed 1040 trials, which were recorded from 44 electrodes at a sampling frequency of 500 Hz and a duration of 4 s. Consistent with prior work [5], we restricted our analysis to 44 electrodes over the sensorimotor belt to emphasize motor-cortical activity.

B. Signal Preprocessing and Segmentation

EEG signals were initially preprocessed to mitigate artifacts. Line interference was suppressed using clean line plug-in in EEGLAB, and ocular, cardiac, and muscular artifacts were eliminated via independent component analysis. A 4-s postcue interval was used per trial. Dataset II was resampled to 250 Hz to match the sampling frequency of dataset I. Signals were decomposed into five frequency subbands, δ (0.5 Hz–3.75 Hz), θ (3.75 Hz–7.5 Hz), α (7.5 Hz–15 Hz), β (15 Hz–30 Hz), and γ (30 Hz–60 Hz) through a 60-Hz low-pass filter followed by wavelet decomposition (using Daubechies mother wavelet). The flowchart of the proposed framework is shown in Fig. 1. Each 4-s signal was then segmented into three overlapping 2-s windows with a sliding interval of 1-s yielding 500 data points for each signal. The graphical overview of the sliding window implementation is shown in Fig. 2. For our analysis, 3888 signals (9 subjects $\times$ 144 trials $\times$ 3 windows) and 4200 signals (14 subjects $\times$ 100 trials $\times$ 3 windows) per class were acquired from datasets I and II, respectively.

III. THEORETICAL OVERVIEW

A. Multilayer Weighted Visibility Graph (WVG)

A visibility graph (VG) maps a univariate time series into an undirected graph where each data point is a node, and edges are formed based on the visibility criterion [12]. The classical VG yields a binary adjacency matrix $A \in \{0, 1\}^{N\times N}$. To capture richer temporal dependencies, a WVG is adopted here, where the edge weight W_{xy} between two nodes (x_a, y_a) and (x_b, y_b) is defined as the inverse of their Euclidean distance, as expressed in the following equation:

$$W_{xy} = \frac{1}{\sqrt{(x_b - x_a)^2 + (y_b - y_a)^2}}. \tag{1}$$

Since EEG signals are multivariate, the WVG is extended to a multilayer configuration, resulting in an MWVG [1]. Consider an M-channel EEG sequence $\{f(t)\}_{t=1}^{N}$ where $f(t) = (f^{[1]}(t), f^{[2]}(t), \ldots, f^{[M]}(t))$. The MWVG comprises M layers, where each layer corresponding to the WVG of a kth layer is the WVG of $\{f^{[k]}(t)\}_{t=1}^{N}$, which represents an individual channel $f^{[k]}(t)$, and adjacency matrix represented as $A^{[k]} = \{a_{ij}{}^{[k]}\}$, where $a_{ij}{}^{[k]} = 1$ if nodes i and j are connected under visibility criterion. To construct rhythm-specific FBC matrices, the mean local efficiency (LE) was calculated for each node in every layer of EEG channels. The LE of all channels were utilized to compute interchannel connectivity through the Pearson correlation coefficient between channel i and j, as defined in the following equation:

$$\rho_{i,j} = \frac{\mathrm{cov}(\mathrm{LE}i,\ \mathrm{LE}j)}{\sigma_{\mathrm{LE}i}\sigma_{\mathrm{LE}j}} \tag{2}$$

where $\rho_{i,j}$ denotes the functional connectivity between channels i and j, $\mathrm{cov}(\cdot)$ is the covariance, and σ is the standard deviation. The process was repeated across five brain rhythms to construct rhythm-specific FBCs, which were then mapped to RGB adjacency image and fed into the proposed lightweight ConvNeXt for automated feature extraction and classification.

B. Proposed Custom ConvNeXt

ConvNeXt is a recent CNN model inspired by Vision Transformer (ViT), designed to capture long-range dependencies while preserving CNN efficiency [13]. Its key innovations include large kernel depthwise convolution of (7×7), layer normalization (LN), Gaussian error linear unit (GELU) activations, and an inverted bottleneck structure. Together, these modifications enhance representational capacity with low computational cost. Given an input feature tensor $x \in \mathbb{R}^{c\times h\times w}$, where c, h, and w represent as channels, height, and width, respectively. The ConvNeXt block is expressed as follows:

$$Y = x + w_b\sigma\left(d_{k\times k}\left(L\left(w_a x\right)\right)\right) \tag{3}$$

where $d_{k\times k}$ represents a depthwise convolution with kernel size $k = 7$, and $w_a \in \mathbb{R}^{4c\times c}$ and $w_b \in \mathbb{R}^{c\times 4c}$ are learnable weight matrices for channel expansion and reduction, respectively. The factor of 4 is a design choice to enrich the intermediate representation, $\sigma(\cdot)$ is the GELU activation, and $L(\cdot)$ denotes LN, expressed as follows:

$$\widehat{z_j} = \frac{z_i - \mu}{\sqrt{\sigma^2 + \psi}} \tag{4}$$

$$\mu = \frac{1}{N}\sum_{j=1}^{N} z_j \tag{5}$$

$$\sigma^2 = \frac{1}{N}\sum_{j=1}^{N}(z_j - \mu)^2 \tag{6}$$

where z_j is activation of the jth feature, $\widehat{z_j}$ is the normalized value, and μ, σ^2, ψ, *and* N representing mean, variance, stability constant, and total number of features respectively. The expansion–compression mechanism forms an inverted bottleneck, enhancing feature diversity while controlling parameters. In this study, a custom ConvNext model with three hierarchical stages was adopted for classifying 224×224×3 FBC images. It uses channel widths of 96, 192, and 384 with three

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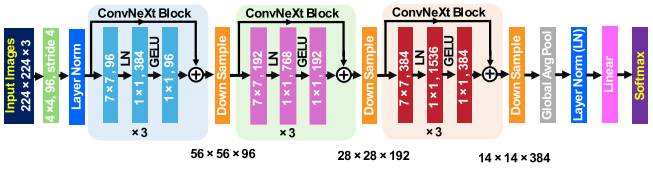


Fig. 3. Architecture of the proposed custom ConvNeXt model.

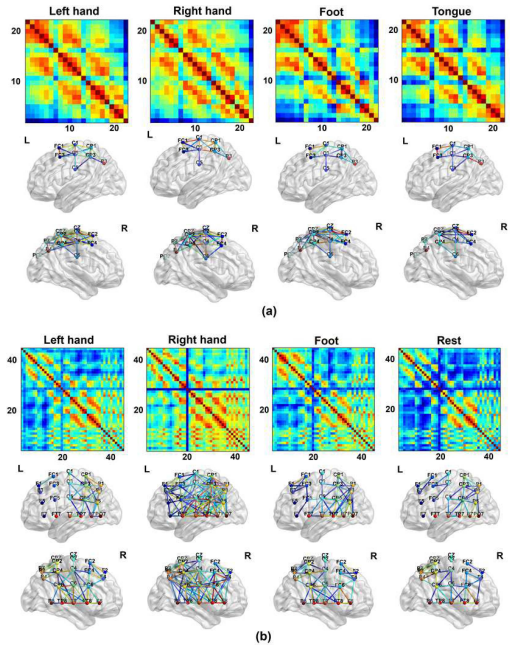


Fig. 4. Channel and regionwise brain connectivity pattern in β and γ bands (Left and Right hemispheres) for (a) dataset I and (b) dataset II.

ConvNeXt blocks per stage, yielding in a lightweight architecture with approximately 5.1 M learnable parameters. Downsampling was performed via 2×2 convolutions with stride 2, reducing the feature resolution to $28 \times 28 \times 192$ and $14 \times 14 \times 384$. After the final stage, global average pooling produced $1 \times 1 \times 384$ descriptor, followed by LN, a linear layer, and a softmax activation for classification. The detailed layerwise architecture is shown in Fig. 3.

IV. RESULTS AND DISCUSSION

A. Analysis of FBC for Different MI Tasks

The rhythm-specific FBC patterns (see Fig. 4) were analyzed by splitting the brain into right (R) and left (L) hemispheres for different MI tasks. In dataset I, the β-band showed strong contralateral sensorimotor coupling, while in dataset II, γ-band revealed strong interhemispheric integration, capturing high-frequency oscillations related to fine motor control. Using an edge-weight threshold ≥ 0.5 emphasized physiologically meaningful connections, and node strength further identifies frontal-central cognitive processes. Regionwise hemisphere-specific lateralization was observed for hand imagery, while foot and tongue tasks showed bilateral engagement. Overall, α, β, and γ rhythms captured discriminative MI-related dynamics, whereas δ and θ are associated with attentional processes. Importantly, β and γ bands directly aligned with the superior classification outcomes, validating the neurophysiological plausibility of the proposed framework.

B. Classification Protocol

To evaluate subject-independent classification task, rhythm-specific FBC matrices were encoded as RGB images and classified through proposed model. For Dataset I, eight subjects were used for training and one for testing, producing 6912 training and 864 testing images. Similarly, for Dataset II, same protocol was adopted for training and testing, yielding 7800 training and 600 testing images. Within each training partition, stratified tenfold cross-validation was performed, splitting 90% for training and 10% for validation, resulting in 6220/ 692 images and 7020/ 780 images (training/ validation) for Datasets I and II, respectively. Each input image, $X \in \mathbb{R}^{224\times224\times3}$ was processed by the proposed custom ConvNeXt to generate pooled feature vector $h \in \mathbb{R}^C$, where C denotes the embedding dimensions. The classifier head computed posterior probabilities for four class MI through the following softmax equation:

$$p(y = j|h) = \frac{\exp\left(w_j^T h + b_j\right)}{\sum_{k=1}^{k} \exp(w_k^T h + b_k)}, j = 1, 2, \cdots, k \quad (7)$$

where $k = 4$ represents the number MI classes and $(w_i,\ b_i)$ are class-specific weight and biases. The categorical cross-entropy objective was minimized through the following equation:

$$L(\emptyset) = -\frac{1}{N} \sum_{i=1}^{N} \sum_{j=1}^{K} y_{ij} \log p(y = j|h_i, \emptyset). \quad (8)$$

Here, N denotes batch size, y_{ij} are labels, and $\emptyset$ is trainable parameter. The optimal hyperparameters were determined through a systematic grid search mechanism. The optimum parameters were taken as learning rate, β_1, β_2 , batch size, dropout rate, L2-regularization, and stochastic depth, which were taken as $\eta = 1 \times 10^{-3}$,0.9, 0.999, 64, 0.5, 0.000014, and 0.1, respectively. Early stopping mechanism and Adam optimizer were adopted to get the best convergence stability consistently. The entire model's performance was assessed through ten iterations to ensure reliability and stability. The whole training was executed on an Intel Core i7 3.6 G Hz CPU, 16 GB RAM, and an NVIDIA RTX 4050 GPU.

C. Classification Performance of the Proposed Model

The performance of the proposed framework was quantitatively evaluated through accuracy (Acc), sensitivity (Sen), specificity (Spe), precision (Pre), and Cohen's kappa coefficient (K), which were derived from confusion matrices. Table 1 summarizes the rhythm specific cross-subject classification results for datasets I and II. In dataset I, the β-band achieved the highest accuracy, followed by the α-band, whereas, in dataset II, the γ-band achieved peak accuracy. Notably, higher frequency bands (α, β, and γ) capture localized oscillatory activity associated with motor cortical engagement, whereas, in contrast, lower frequency δ and θ rhythms yielded weaker performance, reflecting their stronger association with attentional and slow cortical process rather than MI-related neural dynamics.

D. Ablation Studies

The efficacy of the proposed custom ConvNeXt model was compared with benchmark CNNs, such as ResNet18, VGG16, GoogleNet, and ViTs under identical classification protocol and hyperparameter settings as these widely used models are considered as strong baselines for EEG decoding and MI classifications. From Table 2, it is evident that the proposed model achieves superior classification accuracy while requiring substantially fewer parameters and incurring lower computational cost compared to the benchmark architectures. Further, 2-D visualization at the output of the fully connected layers

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TABLE 1. Classification Performance of Datasets I and II

EEG Rhythm	Data-set	Accu (%)	Sens (%)	Spec (%)	Prec (%)	Cohen's kappa
δ	I	78.52 ± 1.3	77.2 ± 1.62	89.5 ± 2.07	77.6 ± 1.93	0.72
	II	79.80 ± 1.72	78.34 ± 1.90	91.56 ± 1.79	78.75 ± 1.83	0.71
θ	I	81.40 ± 0.91	80.53 ± 1.09	91.86 ± 0.96	81.27 ± 1.03	0.72
	II	84.36 ± 0.95	83.75 ± 0.84	93.24 ± 0.88	84.90 ± 0.70	0.76
α	I	84.70 ± 0.82	84.15 ± 0.89	93.63 ± 0.87	84.55 ± 0.82	0.77
	II	88.91 ± 0.77	88.33 ± 0.82	95.14 ± 0.74	88.63 ± 0.86	0.83
β	I	**88.93 ± 0.73**	**87.4 ± 0.74**	**95.0 ± 0.58**	**87.8 ± 0.69**	**0.87**
	II	95.63 ± 0.73	95.15 ± 0.70	98.73 ± 0.71	95.15 ± 0.69	0.91
γ	I	86.37 ± 0.71	86.22 ± 0.70	94.20 ± 0.68	86.23 ± 0.67	0.80
	II	**97.73 ± 0.67**	**98.57 ± 0.60**	**98.3 ± 0.34**	**98.57 ± 0.46**	**0.94**

Bold values indicate best/highest classification performance parameter.

TABLE 2. Comparative Study With Other Benchmark Architectures

Model	Parameters (M)	Accuracy (%)		Computational Time (Minutes)	
		Data set I	Data set II	Data set I	Data set II
ResNet18	11.7	83.49 ± 1.57	91.0 ± 0.93	287	485
VGG16	138.4	81.05 ± 1.45	89.34 ± 0.22	490	850
GoogleNet	6.6	82.36 ± 0.11	90.17 ± 0.82	265	456
ViT	86	84.75 ± 1.06	92.07 ± 0.65	432	706
Custom ConvNeXt	**5.1**	**88.93 ± 0.73**	**97.73 ± 0.67**	**214**	**386**

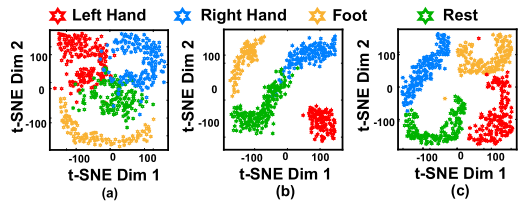


Fig. 5. t-SNE plots of γ-band FBC features for dataset II. (a) ResNet18, (b) ViT, and (c) ConvNeXt model, respectively.

of different CNN models (see Fig. 5) obtained through *t*-stochastic neighbor (t-SNE) further explains superior class separation and high accuracy of the proposed ConvNeXt model.

E. Performance Comparison With Existing Methods

Table 3 shows a comparison of the proposed framework with state-of-the-art MI–EEG methods. The proposed approach consistently outperformed prior models in terms of classification accuracy confirming its reliability for MI decoding using EEG-based BCIs on datasets I and II. Furthermore, the lightweight ConvNext design and cross-subject MI classification demonstrate strong potential for practical, real-time BCI deployment.

TABLE 3. Comparative Study With Other Methods

Paper	Dataset	Method	Accuracy (%)
[2]	I	Attention Temporal CNN	70.97
[3]	II	Graph CNN	80.89
[4]	I	Channel Projection-Mixed CNN	74.6
	II	Channel Projection -Mixed CNN	93.7
[5]	I	Attention Inception + Bi-LSTM	82.8
	II	Attention Inception + Bi-LSTM	97.1
This work	**I**	Custom ConvNeXt	88.93
	II	Custom ConvNeXt	97.73

V. CONCLUSION

This letter presents a novel EEG rhythm-based MI classification framework employing WVG-aided multilayer FBCs and deep learning. FBC matrices for each rhythm were extracted, transformed into RGB images, and then classified using a lightweight custom ConvNeXt model. Experimental results demonstrated that β and γ rhythms yielded the most discriminative patterns, achieving the highest classification accuracy of 90.20% and 96.1%, respectively. Ablation studies and comparisons with state-of-the-art models confirmed superior performance of the proposed model with reduced complexity. However, the proposed framework has been validated only on two datasets. Future work will address cross-dataset generalization, online deployment, and adaptive transfer learning for broader real-time BCI applicability.

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